JOSEPH RIGAL

Robotics Engineer — Embodied AI & Vision-Language-Action Models

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PROFILE

Robotics engineer (Centrale Lyon, 2026; Automation, Robotics & Computer Vision) seeking an AI Scientist / Robotics role from November 2026. Hands-on experience building and deploying learning-based perception and manipulation systems on real robots with LeRobot and vision-language-action (VLA) policies (π0 / π0.5), from sensor integration to a 10× optimisation of a production deep-learning pipeline. Comfortable moving between research prototyping and production-grade ML engineering.

EXPERIENCE

Co-founder | *Robotics & Physical AI start-up — Paris, France* *2026 – present*

R&D Intern — Robotic Perception | *Wormsensing — Grenoble, France* *Apr – Sep 2026 (6 mo)*

- Pioneered a proof of concept for non-visual robotic perception: integrated Wormsensing's Dragonfly® vibration sensor into LeRobot and routed live spectrograms as a "hacked camera" input, enabling a policy to identify objects hidden inside opaque boxes, a task impossible with standard cameras.
- Benchmarked multiple sensing modalities for robot learning (vibration, accelerometers, load cells, strain gauges) to evaluate their fit as alternative perception inputs.
- First author of SpectRobot (arXiv, 2026), with Emmanuel Virot (Wormsensing) and Caroline Pascal (Hugging Face).

Data Scientist Intern | *Probayes — Grenoble, France* *2025 (3 mo)*

- Engineered a 10× speed-up in PDF-to-Markdown conversion for DocParser by profiling and optimising the critical bottlenecks of a deep-learning document-layout-analysis pipeline in production.

Math & Physics Tutor | *Institut Fibonacci — Lyon, France* *2024 – 2026 (2 yrs)*

- Tutored preparatory-class students (Lycée du Parc, Lycée Champollion MPSI) in mathematics and physics.

Assembly Line Intern | *Elistair — Lyon, France* *2023 (4 wks)*

- Assembled tethered drones on the production line.

ROBOTICS PROJECTS & EXTRACURRICULAR

Vision-Language-Action Pick-and-Place Pipeline | *SO-101 arm* *2026*

- Directed a VLA (π0.5) policy by "hacking" the webcam feed: drew placement instructions directly onto the camera stream, a form of visual prompting that let a single model handle boxes of varying sizes and complete long-horizon tasks without retraining.

Hackathon Winner — VIGIE Safety Layer | *Paris Builds × YC Hackathon (Robotics track)* *2026*

- Built VIGIE, an external runtime safety layer for learning-based robots: it watches the robot and workspace through environment cameras, detects dangerous situations, and can pause, hold or resume operation.
- Shipped in 36 hours with a live demo on OpenArm interfaced with LeRobot; the win earned a YC interview with Nicolas Dessaigne for the next batch.

3rd Place — GOSIM Robotics Hackathon | *Paris* *2026*

- Tackled bimanual manipulation with tactile feedback on the OpenArm platform.

Scouting | *Troop Leader* *2023 – 2025*

- Led a scout troop of 32 young people aged 12–17.

EDUCATION

Centrale Lyon — Engineering Degree | *Master's in Automation & Robotics · Deep Learning & CV specialisation* *2023 – 2026*

CPGE Lycée Champollion | *MPI – MP* (Fundamental Mathematics & Physics)* *2021 – 2023*

SKILLS

Robotics / ML: LeRobot, VLA policies (π0 / π0.5), robot learning, embodied AI, sensor fusion, computer vision, model optimisation
Dev: Python (PyTorch), C++, Git
Mechatronics: KiCad, FreeCAD, SolidWorks, tactile sensors
Languages: French (native), English (TOEFL iBT 99/120)

INTERESTS

Trail & mountaineering (since age 11) · Windsurfing · Competitive runner (162nd/352, French Cross-Country Championships; 10 km PB 30:45)